

HONEYWELL

Predictive Quality & Process Reliability in Industrial Operations

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1. THE QUALITY CHALLENGE

Illustrative Case Study — Not Honeywell Internal Data

Examining how industrial organizations manage manufacturing process variation, product and supplier quality variation, automation and system related failures, and process reliability across complex production lifecycles.

Core Problem Statement

“How can critical quality risks be identified early, dominant causes isolated, and process reliability improved before failures recur in industrial environments?”

2. QUALITY ENGINEERING TOOLCHAIN FLOW

1. FISHBONE Possible Causes	2. 5 WHY Drill Down	3. PARETO Prioritise	4. FMEA RPN Risk Assessment	5. RED X Isolate Cause	6. POKA YOKE Error Proof	7. PDCA Improvement	8. PILOT Test Scale	9. SPC Stability	10. CONTROL Sustain
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3. HOW THE TOOLS CONNECT

Tool / Method	Primary Engineering Function	Operational Value
Fishbone and 5 Why	Identify possible causes and drill down to root cause	Structures brainstorming and exposes underlying failure triggers.
Pareto and FMEA RPN	Prioritise major contributors and assess risks (Severity x Occurrence x Detection)	Directs engineering resources toward high-impact system and process failure modes.
Shainin Red X and Poka Yoke	Isolate dominant cause and error-proof the process	Pinpoints exact variation drivers and builds physical and systemic countermeasures.
PDCA, Pilot, SPC and Control	Implement, test, monitor stability and sustain standardization	Ensures engineered solutions permanently control process drift and sustain reliability.

4. POTENTIAL OUTCOMES

- **Earlier identification** of critical quality and process risks.
- **Reduced recurring defects** and unexpected operational failures.
- **Enhanced process reliability** across industrial automation.
- **Improved supplier** and raw material quality control.
- **Minimized process variation** and stronger preventive culture.

KEY TAKEAWAY

“Predictive quality is not only about detecting failures early. It is about identifying variation, isolating dominant causes, and building controls that prevent recurrence permanently.”

Disclaimer: This is an independently created illustrative case study, not an original Honeywell project or representation of Honeywell's proprietary processes, systems or data. Created as part of a 100 Days TQM & Quality Engineering Challenge by Jagriti Singh (Day 59/100).